Certified bounds on the inclusive $\tau \to X_s \nu_\tau$ rate from lattice correlators with a Hausdorff moment condition on $[-1, 1]$

Wei-Lun Chen

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Abstract

Extracting $|V_{us}|$ from the inclusive decay $\tau \to X_s \nu_\tau$ requires an integral of the $\bar{u}s$ spectral function against a known kinematic kernel. The lattice provides only the Euclidean correlator, and inverting it is ill-posed. The Chebyshev method addresses this by assuming that the correlator is a positive combination of λ^t with $\lambda = \mathrm{e}^{-E} \in (0, 1]$. For domain-wall fermions with $M_5 > 1$ the transfer matrix has negative eigenvalues, so this assumption can fail. We find that it does fail on RBC/UKQCD data: the time-component ($J = 0$) correlators cannot be reproduced by a positive measure on $(0, 1]$ within 3σ. They can be reproduced by a positive measure on $[-1, 1]$. We then replace the truncated Chebyshev expansion by a different question: over all positive measures on $[-1, 1]$ that reproduce the data, how large and how small can the kernel integral be? This is a linear program. Its dual is a polynomial that lies above (below) the kernel, and that polynomial certifies the bound independently of the solver. On the $48^3 \times 96$ and $64^3 \times 128$ ensembles, with the HFLAV value $\mathcal{B}_s = 0.02908(48)$, we obtain $0.198 \leq |V_{us}| \leq 0.249$ (0.196–0.251 including the $\mathcal{B}_s$ error). This interval is a guarantee under stated assumptions, not a precision determination.

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1 Introduction

The inclusive width $\Gamma(\tau \to X_s \nu_\tau)$ is proportional to $|V_{us}|^2$ times an integral of the $\bar{u}s$ spectral function $\rho(Q^2)$ against a kernel fixed by kinematics. If ρ were known, $|V_{us}|$ would follow from the measured branching fraction. On the lattice we have the Euclidean correlator $C(t)$, which is the Laplace transform of ρ, and recovering an integral of ρ from noisy values of $C(t)$ is an ill-posed inverse problem.

The approach of Refs. [4, 5, 6] avoids an explicit inversion. One writes $C(t)$ as the expectation of λ^t, $\lambda = \mathrm{e}^{-\hat{H}}$, expands the kernel in shifted Chebyshev polynomials $T_n^*(\lambda)$, and determines the expectations $\langle T_n^* \rangle$ from the data. The key input is that λ^t is averaged over a *positive measure on* $[0, 1]$. That input implies $|\langle T_n^* \rangle| \leq 1$, which keeps the expansion under control.

Applying this method to the RBC/UKQCD $\bar{u}s$ correlators, we ran into three problems:

1. The ensembles use Möbius domain-wall fermions with $M_5 = 1.8$. For $M_5 > 1$ the one-step transfer matrix has real *negative* eigenvalues [2]. A positive measure on $[0, 1]$ is then not guaranteed.
2. In the original implementation the fitted $\langle T_n^* \rangle$ leave $[-1, 1]$ already at $n \simeq 5$–8 (Sec. 6). The resulting $|V_{us}|$ changes by tens of percent from one truncation order to the next.
3. The kernel diverges like $1/E$ at $E \to 0$, and the original code regulates it by hand. Physically no state lies below the kaon, so a lower edge $E \geq M_K$ is available and makes the regulator unnecessary.

This note proposes a remedy that keeps the logic of the calculation but changes two things. First, the support of the measure is enlarged from $(0, 1]$ to $[-1, 1]$, which is what the transfer matrix allows. We test this hypothesis directly on the data with Hausdorff moment conditions. Second, the truncated expansion is replaced by an optimisation. Instead of one estimate with an uncontrolled truncation error, we compute the *largest and smallest* kernel integral compatible with the data and the hypothesis. The answer is an interval, and each end comes with a polynomial certificate that can be checked independently.

Outline. Sec. 1.1 summarises the method and the result on one page. Sec. 2 writes the decay rate as a linear functional of the correlator and fixes all normalisations. Sec. 3 discusses the support of the spectral measure. Sec. 4 is the core of the note and develops the bounds in detail. Sec. 5 tests the hypothesis, Sec. 6 explains why the old Chebyshev coefficients leave $[-1, 1]$, Sec. 7 gives the results, and Sec. 8 states what is and what is not shown.

1.1 The method in one page

1. **Data.** The eight γ structures of the $\bar{u}s$ correlator are grouped into A_T, V_T (temporal, $J = 0$) and A_S, V_S (spatial, $J = 1$). After renormalisation and division by the volume, $C_{\mathrm{ren}} = Z^2 C_{\mathrm{file}}/L^3$ (Eq. 6), the rate is a linear functional of the correlator: $\hat{\Sigma}_{\mathrm{tot}} = \sum_c \int \mathrm{d}\hat{E}\, (2/\hat{E}) K_J \hat{\rho}_c$ and $\Gamma_{\mathrm{app}} = (a^{-1})^2 \hat{\Sigma}$ (Eqs. 7–11).
2. **Hypothesis.** $C(t) = \int \mathrm{d}\sigma(\lambda)\, \lambda^{t-1}$ for $t \geq 1$, with $\sigma \geq 0$ supported on $[-1, \mathrm{e}^{-M_0}]$ and M_0 the kaon mass (Eq. 13). The old method assumes support on $(0, \mathrm{e}^{-M_0}]$.
3. **Constraints.** The moments $m_k = \int \lambda^k \mathrm{d}\sigma$ must lie in $C(k+1) \pm z\delta_{k+1}$ for $k = 0, \ldots, T-1$ (Eq. 15).
4. **Bounds.** The largest and smallest $\int g\, \mathrm{d}\sigma$ over all such σ are solutions of a linear program. Its dual is a polynomial $p \geq g$ (or $q \leq g$) that certifies the bound (Sec. 4). Negative-λ states can absorb part of the moments, but they never contribute to the rate.

5. **Rate.** $|V_{us}| = \sqrt{\mathcal{B}_s \Gamma_\tau / (S_{EW} \frac{G_F^2 m_\tau^3}{16\pi} \Sigma_{\text{tot}})}$, with $\mathcal{B}_s = 0.02908(48)$ from HFLAV 2023 (Sec. 11.6.1 of arXiv:2411.18639).

Result. On the RBC/UKQCD $48^3 \times 96$ and $64^3 \times 128$ ensembles, a positive measure on $(0, 1]$ cannot reproduce the $J = 0$ channels within 3σ (it needs $z^* = 3.3$–19), while one on $[-1, 1]$ can ($z^* \leq 0.1$). The certified interval is $0.198 \leq |V_{us}| \leq 0.249$ (both ensembles, $z = 3$, $T = 18$), or $0.196 \leq |V_{us}| \leq 0.251$ with the $\mathcal{B}_s$ error included. It contains the HFLAV lattice-inclusive value 0.2189(19) and the OPE value 0.2184(21). The interval is about $\pm 11\%$ wide. It is a guarantee, not a precision determination.

Conventions. E is an energy and $\lambda = \mathrm{e}^{-E}$. Lattice quantities are dimensionless and carry a hat where confusion is possible; a^{-1} converts them to GeV. T is the number of correlator values used, $t = 1, \ldots, T$.

2 From the decay rate to a linear functional of the correlator

This section rewrites the decay rate step by step as $\int g \, \mathrm{d}\sigma$ for a known function g. Each step is checked against the single-state case, where the answer is known in closed form.

2.1 The rate

Following Eq. (28) of Ref. [1], with $y = Q^2/m_\tau^2$ and $\Gamma_0 = m_\tau^5 G_F^2/(192\pi^3)$,

$$R \equiv \frac{\Gamma}{\Gamma_0} = 12\pi^2 |V_{us}|^2 S_{EW} \int_0^{m_\tau^2} \frac{\mathrm{d}Q^2}{m_\tau^2} (1-y)^2 \Big[(1+2y)\rho^{(1)}(Q^2) + \rho^{(0)}(Q^2) \Big]. \tag{1}$$

Multiplying by Γ_0 and using $12\pi^2/(192\pi^3) = 1/(16\pi)$ gives

$$\Gamma(\tau \to X_s \nu) = |V_{us}|^2 \, S_{EW} \, \frac{G_F^2 m_\tau^3}{16\pi} \, \Gamma_{\text{app}}, \tag{2}$$

$$\Gamma_{\text{app}} \equiv \int_0^{m_\tau^2} \mathrm{d}Q^2 \Big[K_1(y)\rho^{(1)}(Q^2) + K_0(y)\rho^{(0)}(Q^2) \Big], \tag{3}$$

$$K_1(y) = (1-y)^2(1+2y)\,\theta(1-y), \qquad K_0(y) = (1-y)^2\,\theta(1-y). \tag{4}$$

This is Eq. (44) of Ref. [1], with the measured width $\Gamma_{\text{Exp}} = \mathcal{B}_s \Gamma_\tau$ on the left-hand side.

Check with the kaon pole. For $\rho^{(0)} = f_K^2 \delta(Q^2 - M_K^2)$ we get $\Gamma_{\text{app}} = f_K^2 K_0(M_K^2/m_\tau^2) = f_K^2(1 - M_K^2/m_\tau^2)^2$. This reproduces the textbook width $\Gamma(\tau \to K\nu) = G_F^2 f_K^2 m_\tau^3 (1 - M_K^2/m_\tau^2)^2 |V_{us}|^2/(16\pi)$. For the K^*, $\rho^{(1)} = f_{K^*}^2 \delta(Q^2 - M_{K^*}^2)$ gives $f_{K^*}^2(1-y)^2(1+2y)$, again as expected.

2.2 Spectral representation of the correlator

The zero-momentum correlator is the Laplace transform of the spectral function:

$$C(t) = \int_0^\infty \mathrm{d}Q \, Q^2 \rho(Q^2) \, \mathrm{e}^{-Qt} \;\equiv\; \int_0^\infty \mathrm{d}E \, \rho_E(E) \, \mathrm{e}^{-Et}, \qquad \rho_E(E) = E^2 \rho(E^2). \tag{5}$$

We fix its normalisation with a single state of mass M and decay constant f. Following Eqs. (48) and (51) of Ref. [1], the renormalised correlator per unit volume is

$$C_{\text{ren}}(t) \equiv \frac{Z^2 \, C_{\text{file}}(t)}{L^3} = \frac{M f^2}{2} \, \mathrm{e}^{-Mt}, \qquad f = Z \sqrt{\frac{2A}{ML^3}} \;\; (C_{\text{file}} \simeq A \mathrm{e}^{-Mt}), \tag{6}$$

so $\rho_E = \frac{1}{2} M f^2 \delta(E - M)$. Since $\delta(E - M) = 2M\,\delta(E^2 - M^2)$, this is $\rho(Q^2) = \rho_E/E^2 = f^2\delta(Q^2 - M^2)$, the same normalisation as above. The general case follows by linearity. Here $Z = Z_A = Z_V$.

2.3 The rate in the energy variable

Substituting $Q^2 = E^2$, $\mathrm{d}Q^2 = 2E\,\mathrm{d}E$ and $\rho(E^2) = \rho_E(E)/E^2$ gives

$$\int \mathrm{d}Q^2 K(Q^2)\rho(Q^2) = \int \mathrm{d}E\, 2E\, K\, \frac{\rho_E(E)}{E^2} = \int_0^\infty \mathrm{d}E\, \frac{2}{E}\, K(E)\, \rho_E(E). \tag{7}$$

For the single state, $\rho_E = \frac{1}{2} M f^2 \delta(E - M)$ gives $\frac{2}{M} \cdot \frac{1}{2} M f^2 K = f^2 K$, as it must. The factor $2/E$ is the origin of the divergence at $E \to 0$.

2.4 Lattice units

Let $\hat{E} = aE$, and write the renormalised correlator in lattice units as $c(t) = Z^2 C_{\text{file}}/L^3 = \int \mathrm{d}\hat{E}\, \hat{\rho}(\hat{E}) \mathrm{e}^{-\hat{E}t}$. For a single state $\hat{M} = aM$ and $\hat{f} = af$ in Eq. (6). Defining $\hat{\Sigma} \equiv \int \mathrm{d}\hat{E}\, (2/\hat{E}) K\, \hat{\rho}$,

$$\hat{\Sigma} = \hat{f}^2 K = a^2 f^2 K = a^2\, \Gamma_{\text{app}} \qquad \Longrightarrow \qquad \Gamma_{\text{app}} = (a^{-1})^2\, \hat{\Sigma}, \tag{8}$$

and by linearity this holds for any $\hat{\rho}$. Note that the conversion factor is $(a^{-1})^2$, not $(a^{-1})^3$. The kernel is evaluated at $y = (a^{-1}\hat{E})^2/m_\tau^2$, so its edge sits at $\hat{E} = m_\tau a = 1.027$ (48I) and 0.753 (64I).

2.5 Which components carry $J = 0$ and $J = 1$

Decompose the vacuum polarisation as $\Pi^{\mu\nu}(q) = (q^\mu q^\nu - q^2 g^{\mu\nu})\Pi^{(1)} + q^\mu q^\nu \Pi^{(0)}$. At zero spatial momentum, $q = (q^0, \vec{0})$ and $q^2 = q_0^2$, so

$$\Pi^{00} = (q_0^2 - q^2)\Pi^{(1)} + q_0^2 \Pi^{(0)} = q_0^2\, \Pi^{(0)}, \tag{9}$$

$$\Pi^{ii} = -q^2 g^{ii}\, \Pi^{(1)} = q^2\, \Pi^{(1)} \qquad (\text{no sum over } i,\ g^{ii} = -1). \tag{10}$$

The temporal components ($A_T = \gamma_5\gamma_4$, $V_T = \gamma_4$) therefore carry $J = 0$, and the spatial ones (A_S, V_S, averaged over i) carry $J = 1$:

$$\Sigma_{\text{tot}} = \sum_{X \in \{A,V\}} \left[\Sigma^{(0)}_{X_T} + \Sigma^{(1)}_{X_S}\right], \qquad \Sigma_c^{(J)} = (a^{-1})^2 \int \mathrm{d}\hat{E}\, \frac{2}{\hat{E}} K_J\, \hat{\rho}_c. \tag{11}$$

In the RBC files the gamma labels are $g = 1, 2, 4 \to V_{x,y,z}$, $8 \to V_T$, $14, 13, 11 \to A_{x,y,z}$ and $7 \to A_T$. The original `vus()` assigns $J = 1$ to both axial terms and $J = 0$ to both vector terms. This disagrees with Eq. (11) and with the mock-data code in the same file.

3 The support of the spectral measure

3.1 The standard assumption

If the one-step transfer matrix $\hat{T} = \mathrm{e}^{-\hat{H}}$ is Hermitian and positive, it has eigenvalues $\lambda_n = \mathrm{e}^{-E_n} \in (0, 1]$. With overlaps $c_n = \langle n|J|0\rangle$,

$$C(t) = \sum_n |c_n|^2 \lambda_n^t = \int \mathrm{d}\nu(\lambda)\, \lambda^t, \qquad \nu \geq 0,\ \ \operatorname{supp}\nu \subset (0, 1]. \tag{12}$$

The shifted-Chebyshev method rests on this representation.

3.2 What changes for $M_5 > 1$

Syritsyn and Negele [2] construct $\hat{T}$ for Shamir domain-wall fermions. Its eigenvalues are real, and for $M_5 > 1$ some of them are negative. They do not discuss the signs of the spectral weights. If only link reflection positivity holds, the natural representation is

$$C(t) = \sum_n |c_n|^2 \lambda_n^{t-1} = \int \mathrm{d}\sigma(\lambda)\, \lambda^{t-1}, \qquad t \geq 1, \qquad \sigma = \sum_n |c_n|^2 \delta_{\lambda_n} \geq 0, \quad \operatorname{supp}\sigma \subset [-1, 1]. \tag{13}$$

The moments of σ are the correlator values, $m_k \equiv \int \mathrm{d}\sigma\, \lambda^k = C(k+1)$ for $k = 0, 1, \ldots$. We drop $C(0)$ because it contains contact terms. For $\lambda > 0$ the measure is related to the spectral density by $\mathrm{d}\sigma = \lambda\hat{\rho}\,\mathrm{d}\hat{E}$, since $\int \hat{\rho}\, \mathrm{e}^{-\hat{E}t} \mathrm{d}\hat{E} = \int \lambda^{t-1} (\lambda\hat{\rho}\,\mathrm{d}\hat{E})$. A state with $\lambda < 0$ has positive weight in σ. In the form $C(t) = \int \lambda^t\, \mathrm{d}\nu$ it would have negative weight, since $\mathrm{d}\nu = \mathrm{d}\sigma/\lambda$.

Two hypotheses. We compare $H_{[0,1]}$: $\operatorname{supp}\sigma \subset (0, 1]$ (the old assumption), and $H_{[-1,1]}$: $\operatorname{supp}\sigma \subset [-1, 1]$ (the new one). $H_{[-1,1]}$ is weaker and contains $H_{[0,1]}$. It is *not proven* for Möbius fermions at $M_5 = 1.8$, so we test it on the data in Sec. 5.

Parity of the start time. For the shifted sequence $s_k = C(t_s + k) = \int \mathrm{d}\sigma\, \lambda^{t_s - 1} \lambda^k$ to be the moments of a positive measure on $[-1, 1]$, the factor $\lambda^{t_s - 1}$ must be non-negative there. This requires $t_s - 1$ even, i.e. t_s odd. For even t_s the negative-λ states enter with negative weight, and positivity tests are expected to fail. The old method normalises with $2t_0 = 1$, an odd start time.

4 Certified bounds: the new method

This section is the core of the note. Sec. 4.1 states the idea without formulas. Secs. 4.2–4.3 set up the optimisation and derive the certificate. Sec. 4.4 works the method out by hand for two moments. Sec. 4.5 describes what the code does and which step makes the result rigorous. Sec. 4.6 shows what the extremal spectra look like on the data, and Sec. 4.8 collects the steps.

4.1 The idea

The data fix finitely many numbers, $C(1), \ldots, C(T)$, each with an error. Many spectral measures reproduce them, and these measures give different values of the kernel integral $\hat{\Sigma}$. The Chebyshev method picks *one* of them, through a truncated expansion and a prior, and quotes a central value whose truncation error is not controlled. We ask a different question: *over all positive measures on the allowed support that reproduce the data within their errors, what are the largest and the smallest possible* $\hat{\Sigma}$*?* The answer is an interval that contains the true value whenever the hypotheses hold. No truncation, prior or basis enters.

Two facts make this question tractable. First, $\hat{\Sigma}$ and the moments are both *linear* in the measure, so finding the extremes is a linear program (LP). Second, every LP has a dual problem. Here the dual variables assemble into a polynomial $p(\lambda)$ that lies above the kernel. This polynomial can be checked after the fact: if $p \geq g$ on the support, the upper bound holds, regardless of how the solver arrived at it.

4.2 The optimisation problem

For a channel c with angular momentum J, define $\kappa_J(\lambda) = \frac{2}{E} K_J\big((a^{-1}E)^2/m_\tau^2\big)$ with $E = -\ln\lambda$. Combining Eq. (11) with $\mathrm{d}\sigma = \lambda\hat{\rho}\,\mathrm{d}\hat{E}$ gives

$$\hat{\Sigma}_c = \int \mathrm{d}\sigma(\lambda)\, g(\lambda), \qquad g(\lambda) = \begin{cases} \kappa_J(\lambda)/\lambda, & 0 < \lambda \le \mathrm{e}^{-M_0}, \\ 0, & \lambda \le 0. \end{cases} \tag{14}$$

The function g (black curve in Fig. 1) has three regions, and it helps to keep them apart.

- *The kernel window,* $\lambda_c < \lambda \le \mathrm{e}^{-M_0}$, i.e. $M_0 \le E < m_\tau$, with $\lambda_c = \mathrm{e}^{-am_\tau}$ (0.358 on 48I, 0.471 on 64I). Only here is $g > 0$. It rises steeply towards the kaon; for $J = 0$ on 48I, $g(\mathrm{e}^{-M_0}) = 7.86$.
- *Heavy states,* $0 \le \lambda \le \lambda_c$, i.e. $E \ge m_\tau$. These states are kinematically closed, so $g = 0$, but they still contribute to $C(t)$.
- *Negative eigenvalues,* $-1 \le \lambda < 0$. These states of the transfer matrix (Sec. 3) have no physical energy, and we set $g = 0$ for them. This is an assumption: we treat them as lattice artefacts that are absent from the continuum spectral function.

States in the last two regions *take up part of the moments without contributing to the rate.* This is the mechanism by which the data constrain $\hat{\Sigma}$ from below as well as from above. The upper edge e^{-M_0}, given by the kaon as the lightest $\bar{u}s$ state, is needed because $g \sim 2/E$ diverges as $E \to 0$. Without it the upper bound would be infinite.

We seek bounds over all positive measures that are consistent with the data:

$$\hat{\Sigma}^{\pm} = \ \sup / \inf \int g\,\mathrm{d}\sigma \quad \text{s.t.} \quad \sigma \ge 0 \text{ on } S, \quad |m_k - C(k+1)| \le z\,\delta_{k+1}, \quad k = 0, \ldots, T-1. \tag{15}$$

Here $S = [-1, \mathrm{e}^{-M_0}]$ (new) or $(0, \mathrm{e}^{-M_0}]$ (old), $m_k = \int \lambda^k \mathrm{d}\sigma$, and δ_t is the jackknife error of $C(t)$. The constraints are T "boxes", one per correlator value, each with a half-width of z standard deviations.

Enlarging the support can only widen the interval. Every measure on $(0, \mathrm{e}^{-M_0}]$ is also a measure on $[-1, \mathrm{e}^{-M_0}]$. The new feasible set therefore contains the old one, and

$$[\hat{\Sigma}^-_{\text{old}}, \hat{\Sigma}^+_{\text{old}}] \subseteq [\hat{\Sigma}^-_{\text{new}}, \hat{\Sigma}^+_{\text{new}}] \qquad \text{whenever the old problem is feasible.} \tag{16}$$

The new hypothesis is weaker and therefore safer. On our data the cost is small. For $J = 0$ the old problem has no solution at all, and for $J = 1$ its bounds are only slightly tighter (Table 3).

4.3 The dual: where the certificate comes from

Write $C_k \equiv C(k+1)$ and $\delta_k \equiv \delta_{k+1}$. For the upper bound, attach a multiplier $y_k^+ \ge 0$ to $m_k \le C_k + z\delta_k$ and $y_k^- \ge 0$ to $-m_k \le -(C_k - z\delta_k)$. For every feasible σ and every choice $y^\pm \ge 0$, the added terms are non-negative, so

$$\begin{aligned} \int g\,\mathrm{d}\sigma \ &\le \ \int g\,\mathrm{d}\sigma + \sum_k y_k^+ (C_k + z\delta_k - m_k) + \sum_k y_k^- (m_k - C_k + z\delta_k) \\ &= \int (g - p)\mathrm{d}\sigma + \sum_k \left[p_k C_k + z\delta_k (y_k^+ + y_k^-)\right], \end{aligned} \tag{17}$$

where $p_k \equiv y_k^+ - y_k^-$ and $p(\lambda) = \sum_k p_k \lambda^k$. If $p \ge g$ on S, the first term is ≤ 0 because $\sigma \ge 0$. It is optimal to make at most one of $y_k^\pm$ non-zero, in which case $y_k^+ + y_k^- = |p_k|$. This proves:

Proposition 1. *Let $p(\lambda) = \sum_{k=0}^{T-1} p_k \lambda^k$ satisfy $p \geq g$ on S. Then every σ that is feasible in (15) satisfies*

$$\int g \, \mathrm{d}\sigma \;\leq\; U(p) \equiv \sum_k p_k C(k+1) + z \sum_k |p_k| \, \delta_{k+1}. \tag{18}$$

Likewise, if $q \leq g$ on S, then $\int g \, \mathrm{d}\sigma \geq L(q) \equiv \sum_k q_k C(k+1) - z \sum_k |q_k| \, \delta_{k+1}$.

The best certificate minimises $U(p)$ over all polynomials with $p \geq g$ on S. This is the *dual* of (15). For the moment problem on a compact interval there is no duality gap, so $\min_p U(p) = \hat{\Sigma}^+$. Three properties of the dual help in reading the results.

- *Sensitivity.* $p_k = \partial \hat{\Sigma}^+ / \partial C(k+1)$, so the certificate tells how much the bound moves when one data point moves. The term $z|p_k|\delta_{k+1}$ is the price paid for the error on $C(k+1)$. A polynomial with large alternating coefficients is penalised. This is the noise amplification of Sec. 6 seen from the other side.
- *Complementary slackness.* At the optimum, the maximising σ puts weight only where $p(\lambda) = g(\lambda)$. Also, $y_k^+ > 0$ only if m_k sits at the upper edge of its box. In Fig. 1 each polynomial touches g exactly at the atoms of the corresponding extremal measure.
- *Role of the support.* The condition $p \geq g$ is imposed only on S. Under the old support, p is unrestricted for $\lambda < 0$. Under the new support, $p \geq 0$ is also required on $[-1, 0)$. More constraints give a larger minimum. This is the dual form of the inclusion above.

If g were a polynomial, the condition $p - g \geq 0$ on $[-1, 1]$ would be equivalent to $p - g = s_0 + (1 - \lambda^2) s_1$ with sums of squares s_0, s_1 (Lukács). That turns the dual into a semidefinite program, and it is the grid-free version listed under next steps in Sec. 8.

4.4 A worked example with two moments

To see how the bounds arise, take $T = 2$ and exact data ($z\delta \to 0$). Let $m_0 = C(1)$, and let $\mu = C(2)/C(1)$ be the mean of λ under the normalised measure σ/m_0. The certificate is now a straight line, $p(\lambda) = a + b\lambda$, and $U = m_0(a + b\mu)$ is m_0 times its value at $\lambda = \mu$. The lowest line that stays above g on S, evaluated at μ, is the *concave envelope* $\hat{g}_S(\mu)$. Similarly, the highest line below g gives the convex envelope $\check{g}_S(\mu)$. Therefore

$$m_0 \, \check{g}_S(\mu) \;\leq\; \hat{\Sigma} \;\leq\; m_0 \, \hat{g}_S(\mu). \tag{19}$$

For the A_T channel on 48I we have $\mu = 0.104$, $\mathrm{e}^{-M_0} = 0.750$ and $g(\mathrm{e}^{-M_0}) = 7.86$.

- *Lower bound.* Since $\mu < \lambda_c = 0.358$, a single atom at $\lambda = \mu$, i.e. one state above m_τ, reproduces both moments and contributes nothing to the rate. Hence $\hat{\Sigma}^- = 0$: two moments cannot tell that a kaon is present.
- *Upper bound.* g vanishes left of λ_c and is convex inside the window, so $\hat{g}_S$ is the chord from the left end of S to the point $(\mathrm{e}^{-M_0}, g(\mathrm{e}^{-M_0}))$. The maximising measure is an atom at the kaon plus an atom at the left end of S. With the old support the left end is $\lambda = 0$, and $\hat{\Sigma}^+ = m_0 \, g(\mathrm{e}^{-M_0}) \, \mu / \mathrm{e}^{-M_0} = 0.042 \, \mathrm{GeV}^2$. With the new support it is $\lambda = -1$, and $\hat{\Sigma}^+ = m_0 \, g(\mathrm{e}^{-M_0})(1 + \mu)/(1 + \mathrm{e}^{-M_0}) = 0.189 \, \mathrm{GeV}^2$. An atom at $\lambda = -1$ lowers m_1, which leaves room for more weight on the kaon.

The LP reproduces both numbers to five digits. Adding moments closes the interval quickly. With the new support and $z = 3$:

T	2	4	6	8	12	18
$\hat{\Sigma}_{A_T}$ [GeV^2]	[0, 0.189]	[0.0177, 0.0232]	[0.0203, 0.0214]	[0.0205, 0.0212]	[0.0205, 0.0212]	[0.0205, 0.0212]

The reason is visible in the effective mass of A_T: 2.26, 0.77, 0.46, 0.32, 0.30, 0.29 at $t = 1, \ldots, 6$. The kaon ($aM_K = 0.288$) dominates from $t \approx 5$. Once the moments reach that far, the weight at the kaon is pinned, and the interval closes to $\pm 2\%$ around the kaon-pole value 0.0208. With the old support the same problem has no solution from $T = 6$ on.

4.5 Numerics, and which step is rigorous

The linear program. Because g is not a polynomial, we solve the primal problem on a grid $\{\lambda_i\}$ (function `make_grid`). The grid has 3000 points uniform in $E \in [M_0, 3]$, 1000 points in $[3, 9]$, the points $E = 14$ and 40, and $\lambda = 0$. For the new support it also has 900 points in $[-1, -10^{-4}]$, uniform down to -0.05 and geometric below. The measure is $\sigma = \sum_i w_i \delta_{\lambda_i}$. We use variables $u_i = w_i / C(1) \geq 0$ and divide each box by δ_{k+1}, so all rows are of order one:

$$\max / \min \sum_i C(1) g(\lambda_i)\, u_i \quad \text{s.t.} \quad \Big| \sum_i \frac{C(1)\lambda_i^k}{\delta_{k+1}} u_i - \frac{C(k+1)}{\delta_{k+1}} \Big| \leq z, \quad k = 0, \ldots, T-1. \tag{20}$$

The LP is solved with HiGHS through `scipy.optimize.linprog`. Restricting σ to a grid can only shrink the feasible set, so the grid bounds are an *inner* approximation: they can be too narrow, never too wide. The next step repairs this.

The extremal measures have few atoms. A basic optimal solution has at most as many non-zero u_i as there are active constraints. Since $z > 0$, the two sides of a box cannot both be active. *The extremal measures therefore have at most T atoms.* On the data they have 5–8. This is why Fig. 1 shows a few isolated spikes; adjacent grid points occasionally share one atom.

From the grid to a certificate. The following steps make the bound rigorous (function `certified_bounds`):

1. Read the dual values $y_k^{\pm}$ of the two rows of box k from the solver and set $p_k = (y_k^+ - y_k^-)/\delta_{k+1}$, which undoes the row scaling. By LP duality, $p \geq g$ holds at every grid point.
2. Evaluate $p - g$ on a much denser grid of the whole support: 2×10^5 points uniform in λ over the positive part, the point $\lambda = 0$, and 5×10^4 points in $[-1, 0)$.
3. Set $\Delta = \max(0, \max(g - p))$ and shift $p_0 \to p_0 + \Delta$. Now $p \geq g$ holds on the dense grid.
4. Evaluate $U(p)$ from Eq. (18). For the lower bound, repeat with q, $\Delta = \max(0, \max(q - g))$, $q_0 \to q_0 - \Delta$, and $L(q)$.

The shift moves the bound by about $\Delta \cdot C(1)$. On the data the certified and grid bounds differ by at most $3 \times 10^{-5}\,\mathrm{GeV}^2$ (Table 3). The dense-grid check is numerical, not an analytic proof that $p \geq g$. However, g is smooth in each region and p is a polynomial of degree at most 17, so a violation would have to fit between grid points spaced by 3×10^{-6}. The sum-of-squares formulation above would remove this caveat.

Statistical meaning. Each box has half-width $z\delta_t$. With Gaussian errors, all T boxes contain the true values with probability at least $1 - T \cdot 0.0027$ for $z = 3$ (Bonferroni), i.e. at least 95% for $T = 18$. The covariance between different t is not used. Including it, as an ellipsoid instead of a box, would shrink the feasible set and tighten the bounds.

Choice of T. At fixed z, each additional moment adds a constraint. The interval can therefore only shrink as T grows, and the bound is valid for every T. Once $\delta_t / C(t)$ becomes large, the new box is too wide to ever be active, and the bounds stop improving (Fig. 3). We use $T = 18$. The only cost of a larger T is the Bonferroni factor.

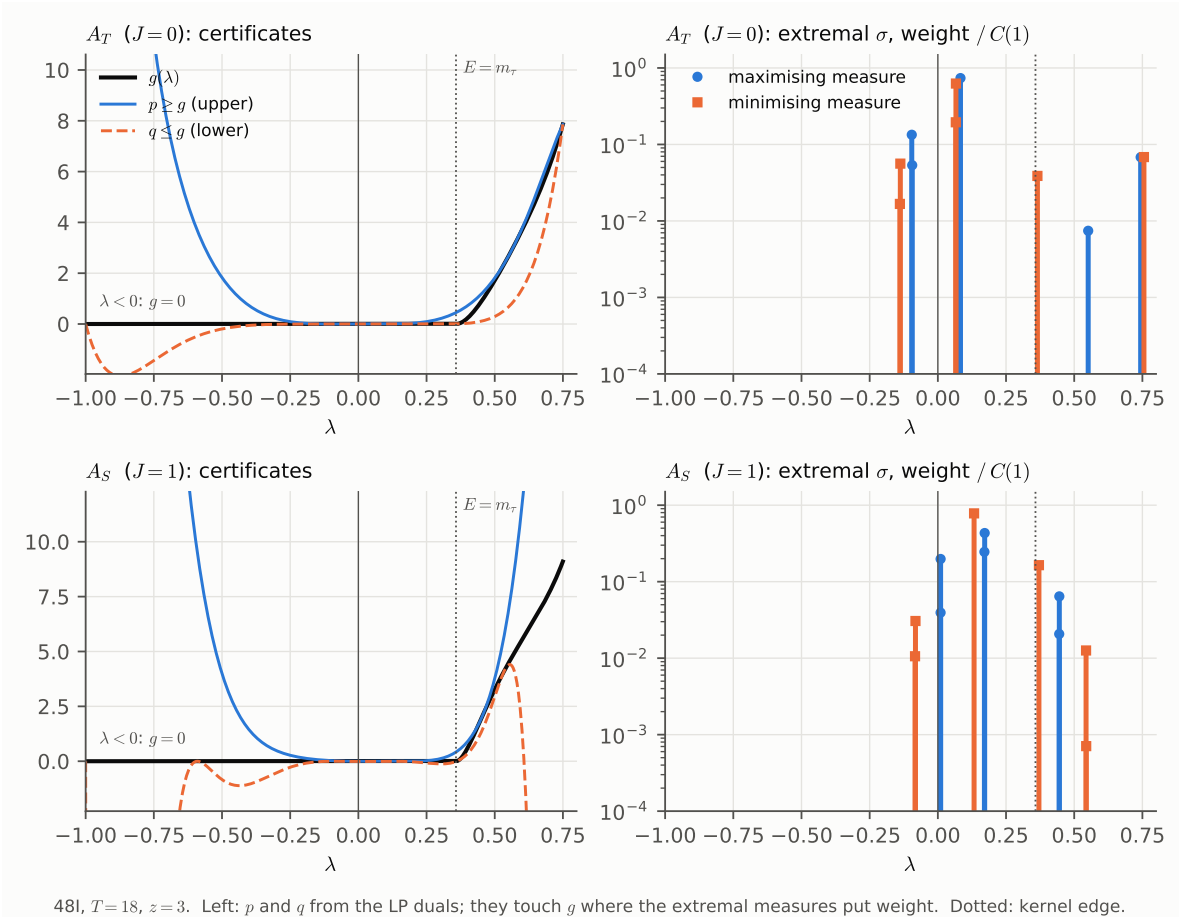


Figure 1: 48I, $T = 18$, $z = 3$, new support. *Left:* the kernel $g(\lambda)$ (black), the upper certificate $p \geq g$ (blue) and the lower certificate $q \leq g$ (orange, dashed), read from the LP duals. *Right:* the measures that attain the upper (blue) and lower (orange) bound, with weights relative to $C(1)$ on a log scale. Each certificate touches g where its measure has an atom. The dotted line is the kernel edge $E = m_\tau$. Everything to its left contributes to $C(t)$ but not to Σ.

Calibration. For a single state ($m = 0.4937\,\text{GeV}$, $f = 0.1564\,\text{GeV}$ on 48I, relative errors 10^{-3}) the method gives $\Sigma \in [0.02075, 0.02089]\,\text{GeV}^2$ for both supports, around the exact value $f^2(1 - M^2/m_\tau^2)^2 = 0.02083$. The certificate reproduces the grid bound to five digits.

4.6 What the extremal spectra look like

Fig. 1 shows the most and least precise channels.

- A_T *($J = 0$, interval ±1.7%).* Both extremal measures have an atom at the kaon, $\lambda = \mathrm{e}^{-M_0} = 0.750$, with weight $0.068\,C(1)$. It supplies 96–100% of Σ. The rest of the weight lies outside the window: at $\lambda \approx 0.06$–0.09 ($\hat{E} \approx 2.4$–2.8, far above m_τ) and at $\lambda \approx -0.09$ to -0.15, where 7–19% of the weight sits. *The data cannot be described without these negative-λ atoms*: removing them, which is the old support, makes the problem infeasible at $z = 3$ from $T = 6$ on (Sec. 4.7). The upper and lower bounds differ only through a small amount of weight inside the window. The maximiser puts 0.7% at $\lambda = 0.56$, while the minimiser puts 3.9% just outside the edge, at $\lambda = 0.359$.
- A_S *($J = 1$, interval ±48%).* There is no isolated low-lying state. The maximiser places 8.5% of the weight at $\lambda = 0.45$ ($E = 1.37\,\text{GeV}$). The minimiser places 1.3% at $\lambda = 0.54$ and 16% just outside the edge. Both are in the K^*–K_1 region. Because these correlators decay quickly, the data do not determine *where* inside the window the weight sits. The relative error of $C(t)$

is 3% at $t = 10$ and 185% at $t = 18$. Since g changes by a factor of several across the window, so does Σ. This channel limits the $|V_{us}|$ interval.

The certificates themselves are ordinary polynomials of degree 17. They follow g closely near the atoms and are free elsewhere. For A_T, p must stay non-negative on $[-1, 0)$ and rises steeply towards $\lambda = -1$. This is where the new support costs something, and the cost is small because the data leave almost no room for weight there.

4.7 Feasibility

Before computing bounds we ask whether any positive measure on S reproduces the data at all. The smallest tolerance z^* for which this is possible is again an LP: minimise $\max_t |\sum_i w_i \lambda_i^{t-1} - C(t)|/\delta_t$ over $w \geq 0$. At $z = 3$ the old support is therefore *infeasible* for A_T and V_T, at both

z^* at $T = 14$ (old $[0,1]$ / new $[-1,1]$)	A_T	V_T	A_S	V_S
48I	3.28 / 0.09	5.22 / 0.07	0.64 / 0.39	0.27 / 0.20
64I	9.06 / 0.06	18.72 / 0.08	0.75 / 0.10	0.78 / 0.05

Table 1: Minimal tolerance (in units of σ) for the data to be reproduced by a positive measure. For $T \geq 6$ the $J = 0$ channels need $z^* > 3$ with the old support.

lattice spacings and for every $T \geq 6$. This is the moment-problem version of the observation that "the Chebyshev coefficients exceed 1". The test uses the same LP and needs no input about $|V_{us}|$.

4.8 The recipe

1. For each channel, build $C_{\rm ren}(t)$ (Eq. 6) and its jackknife error δ_t for $t = 1, \ldots, T$.
2. Build the grid on $S = [-1, \mathrm{e}^{-M_0}]$ and evaluate $g(\lambda)$ from Eq. (14) for the channel's J.
3. Compute z^*. If $z^* > z$, the hypothesis is rejected for this channel and no bound is computed.
4. Solve the two LPs (15) and read off p and q from the dual values.
5. Check $p \geq g \geq q$ on the dense grid, shift by Δ, and compute $U(p)$ and $L(q)$. These are $\hat{\Sigma}_c^{\pm}$.
6. Add the four channels: $\Sigma_{\rm tot}^{\pm} = (a^{-1})^2 \sum_c \hat{\Sigma}_c^{\pm}$. This is conservative, because each channel may sit at its own extreme.
7. Convert to $|V_{us}| \in [\sqrt{c/\Sigma_{\rm tot}^{+}}, \sqrt{c/\Sigma_{\rm tot}^{-}}]$ with $c = \mathcal{B}_s \Gamma_\tau / (S_{EW} G_F^2 m_\tau^3 / 16\pi)$ (Sec. 7.2).

5 Testing the hypothesis: Hausdorff positivity

The feasibility test of Sec. 4.7 uses all moments together. This section gives an independent and more local test, based on the classical characterisation of moment sequences.

5.1 Necessary conditions

Let $P(\lambda) = \sum_{i=0}^{n} x_i \lambda^i$ be a real polynomial and $s_k = \int \mathrm{d}\sigma\, \lambda^k\, \lambda^{t_s - 1}$. If $\sigma \geq 0$ on $[-1, 1]$ and t_s is odd, then

$$x^\top H x = \sum_{ij} x_i x_j\, s_{i+j} = \int \mathrm{d}\sigma\, \lambda^{t_s-1} P(\lambda)^2 \;\geq 0, \qquad H_{ij} = s_{i+j}, \tag{21}$$

$$x^\top H^{\rm loc} x = \int \mathrm{d}\sigma\, \lambda^{t_s-1}(1-\lambda^2) P(\lambda)^2 \;\geq 0, \qquad H^{\rm loc}_{ij} = s_{i+j} - s_{i+j+2}. \tag{22}$$

The Hankel matrix H and the localizing matrix H^{loc} must therefore be positive semidefinite. For support $[0,1]$ the localizing matrix is instead $H^{\mathrm{loc}}_{ij} = s_{i+j+1} - s_{i+j+2}$, from the weight $\lambda(1-\lambda)$. Schmüdgen [3] shows that for $s_0, \ldots, s_{2n}$ these conditions are also *sufficient*, with no rank condition on a compact interval. Here only necessity matters. A significantly negative eigenvalue *excludes* the hypothesis, while passing the test means compatibility, not proof.

5.2 Scaling and statistics

The values s_k decay by orders of magnitude, so we scale $H \to D^{-1}HD^{-1}$ with $D = \mathrm{diag}(\sqrt{|H_{ii}|})$. This is a congruence, so by Sylvester's law it does not change positive semidefiniteness. For each (t_s, n) we compute the smallest eigenvalue $\mu_{\min}$ and its jackknife error over configurations. We call a matrix a *violation* if $\mu_{\min}/\sigma < -3$ and a *pass* if $\mu_{\min}/\sigma > 3$. The rows of the data files carry no configuration label. The autocorrelation along the row order is at the noise level ($|\mathrm{ac}| < 0.05$ for lags 1–100), so the jackknife error does not depend on the assumed ordering.

5.3 Result

channel	support	odd t_s (pass/inc/viol)	even t_s
V_S, A_S ($J=1$), 48I and 64I	$[-1,1]$	5–7 / 1–3 / 0	4–6 / 2–4 / 0
V_T ($J=0$), 64I	$[-1,1]$	5/3/0	2/3/**3**
V_T ($J=0$), 64I	$[0,1]$	0/5/**3**	3/5/0
A_T ($J=0$), 64I	$[-1,1]$	5/3/0	3/3/2
A_T ($J=0$), 64I	$[0,1]$	3/3/**2**	3/5/0

Table 2: Localizing matrices with $n \geq 2$ and $t_s \leq 8$. At odd t_s the support $[-1,1]$ is never violated, in any channel or ensemble. The $J=0$ channels show the pattern expected from negative-λ states. The full table is produced by `hankel_scan.py`.

The pattern in Table 2 is the one predicted by Eq. (13). The $J=0$ channels violate $[0,1]$ at odd t_s, where negative-λ states have positive weight that a $[0,1]$ measure cannot mimic. They violate $[-1,1]$ only at even t_s, where those states enter with negative weight. The $J=1$ channels pass every test and do not discriminate between the hypotheses. Because the same data enter many (t_s, n) tests, isolated 3σ violations can occur by chance. V_T is also the least clean channel, since the local vector current is not conserved for $m_u \neq m_s$.

6 Why the old Chebyshev coefficients leave $[-1,1]$

6.1 Definitions and the old relation

With $\lambda = (1+\cos\theta)/2 = \cos^2(\theta/2)$, the shifted Chebyshev polynomials are $T^*_j(\lambda) = \cos j\theta = T_j(2\lambda - 1)$. Normalise $\bar{C}(t) = C(t+2t_0)/C(2t_0) = m_t/m_0 \equiv \langle \lambda^t \rangle$. Expanding λ^t in Fourier modes,

$$\lambda^t = \cos^{2t}\tfrac{\theta}{2} = 2^{-2t}\sum_{k=0}^{2t}\binom{2t}{k}\mathrm{e}^{\mathrm{i}(t-k)\theta} = 2^{1-2t}\Big[\tfrac{1}{2}\binom{2t}{t} + \sum_{j=1}^{t}\binom{2t}{t-j}\cos j\theta\Big], \tag{23}$$

$$\bar{C}(t) = 2^{1-2t}\Big[\tfrac{1}{2}\binom{2t}{t} + \sum_{j=1}^{t}\binom{2t}{t-j}\langle T^*_j\rangle\Big]. \tag{24}$$

This is Eq. (56) of Ref. [1], with $\langle T^*_0\rangle = 1$ by normalisation. The old method fits the $\langle T^*_j\rangle$ to $\bar{C}(t)$, $t = 1, \ldots, T$, with the prior $|\langle T^*_j\rangle| < 1$ imposed through an erfinv map.

6.2 The bound, and how negative eigenvalues break it

Lemma 1. *If $\sigma \geq 0$ is normalised on $[0,1]$, then $|\langle T_n^* \rangle| \leq 1$. If $\sigma \geq 0$ is normalised on $[-1,1]$, then $|\langle T_n \rangle| \leq 1$.*

Proof. $|T_n^*| \leq 1$ on $[0,1]$ and $|T_n| \leq 1$ on $[-1,1]$, so $|\langle T \rangle| \leq \int |T| \, \mathrm{d}\sigma \leq \int \mathrm{d}\sigma = 1$. □

Now place a small weight w at $\lambda = -\varepsilon$ with $\varepsilon > 0$. Since $2\lambda - 1 = -1 - 2\varepsilon$ lies outside $[-1,1]$, the Chebyshev polynomial grows exponentially there:

$$T_n^*(-\varepsilon) = (-1)^n \cosh\bigl(n \operatorname{arccosh}(1+2\varepsilon)\bigr) \simeq \tfrac{1}{2}(-1)^n \mathrm{e}^{\,n\kappa}, \quad \kappa = \operatorname{arccosh}(1+2\varepsilon). \tag{25}$$

The contribution $w\,T_n^*(-\varepsilon)$ alone exceeds 1 once $n \gtrsim \ln(2/w)/\kappa$. For $\varepsilon = 0.1$ ($\kappa = 0.62$) this happens at $n \gtrsim 8.5$ for $w = 1\%$ and at $n \gtrsim 4.8$ for $w = 10\%$. A small negative-λ component is therefore enough to push the old coefficients out of $[-1,1]$ at the orders observed in the data. The extremal measures of Sec. 4.6 have exactly such components, at $\lambda \approx -0.1$ with 7–19% of the weight.

6.3 Noise amplification

The monomial coefficients c_k of $T_n^*(x)$ alternate in sign, because all roots lie in $(0,1)$ (Descartes' rule). Hence $\sum_k |c_k| = |T_n^*(-1)| = T_n(3) \simeq \frac{1}{2}(3+2\sqrt{2})^n$. The coefficients of $T_n(x)$ alternate in x^2, so $\sum_k |c_k| = |T_n(\mathrm{i})| \simeq \frac{1}{2}(1+\sqrt{2})^n$. At $n = 12$ these sums are 7.7×10^8 and 2.0×10^4. The $[-1,1]$ basis therefore amplifies the statistical noise of the moments about 4×10^4 times less. The price is a coarser resolution in λ, roughly a factor of two in polynomial degree.

6.4 Data

7 Results

7.1 Channel bounds

$\hat{\Sigma}$ [GeV^2], $T = 18$, $z = 3$	A_T	V_T	A_S	V_S
48I new $[-1,1]$	[0.02048, 0.02119]	[0.00039, 0.00135]	[0.01021, 0.02894]	[0.03764, 0.05907]
64I new $[-1,1]$	[0.02048, 0.02125]	[0.00038, 0.00127]	[0.00617, 0.02761]	[0.03653, 0.05876]
old $[0,1]$	infeasible		feasible, slightly tighter lower bounds	

Table 3: Certified bounds per channel ($T = 18$, $z = 3$). The kaon pole alone gives $f_K^2(1 - M_K^2/m_\tau^2)^2 = 0.02084$, inside the A_T interval on both ensembles. This confirms the normalisation of Sec. 2 on real data. Certified and grid bounds differ by at most $3 \times 10^{-5}\,\mathrm{GeV}^2$.

Table 3 lists the bounds per channel. The two lattice spacings agree within the intervals. The A_T channel is fixed to ±2% by the kaon, and the two $J = 1$ channels dominate the width.

7.2 $|V_{us}|$

Eq. (44) of Ref. [1] with $\Gamma_{\mathrm{Exp}} = \mathcal{B}_s \Gamma_\tau$ gives

$$|V_{us}| = \sqrt{\frac{\mathcal{B}_s\,\Gamma_\tau}{S_{EW}\,\dfrac{G_F^2 m_\tau^3}{16\pi}\,\Sigma_{\mathrm{tot}}}}, \qquad \Sigma_{\mathrm{tot}} = \sum_c \Sigma_c^{\pm}. \tag{26}$$

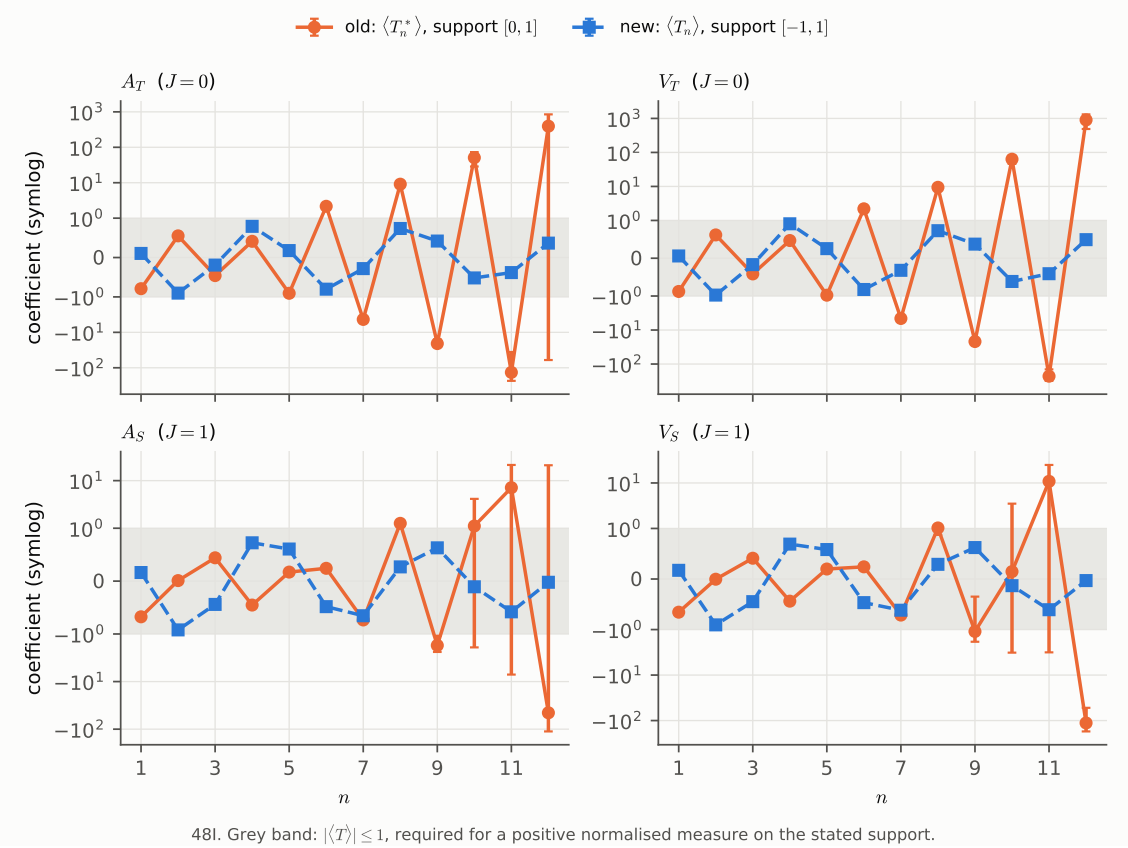


Figure 2: $\langle T_n^* \rangle$ (old, orange) and $\langle T_n \rangle$ (new, blue) on 48I for the four channels, with jackknife errors, on a symmetric-log axis. The grey band is $[-1, 1]$. The old coefficients leave the band at $n = 5$–8 (e.g. A_T, $n = 6$: 2.16 ± 0.04) and reach 10^2–10^3 at $n = 12$. The new ones stay inside for all channels and $n \leq 12$ (maximum 0.98, error ≤ 0.02). They follow $\cos(n\pi/2)$ because the measure is concentrated near $\lambda = 0$, so this is a necessary but weak test.

The inputs are $\mathcal{B}_s = 0.02908 \pm 0.00048$ (HFLAV 2023, arXiv:2411.18639, Sec. 11.6.1, Eq. (246); the fit table gives $\mathcal{B}(\tau^- \to X_s^- \nu_\tau) = 0.0291(5)$), $\Gamma_\tau = \hbar/\tau_\tau = 2.267 \times 10^{-12}\,\text{GeV}$, $S_{EW} = 1.0201$ and $G_F^2 m_\tau^3/(16\pi) = 1.518 \times 10^{-11}\,\text{GeV}^{-1}$. We add the bounds of the four channels, which is conservative since the channels are bounded independently. Then $|V_{us}| \in [\sqrt{c/\Sigma_{\text{tot}}^+}, \sqrt{c/\Sigma_{\text{tot}}^-}]$ with $c = \mathcal{B}_s \Gamma_\tau / (S_{EW} G_F^2 m_\tau^3/16\pi)$.

7.3 Comparison with the original code

Figure 4 compares the certified interval with the values from the original code, run unchanged except for the removal of `cupy`, for both kernel assignments discussed in Sec. 2. The old values move by up to $\pm 30\%$ between consecutive T, while their quoted errors ($\approx 6 \times 10^{-4}$) do not include the truncation error. Several points in the old code should be checked before its numbers are used (details in `OLD_CODE_NOTES.md`): the kernel assignment, the ignored `-t0` argument (it lands in `ND`), the absence of $\mathcal{B}_s$ from the $|V_{us}|$ formula, and the position of Z_A^2, which should multiply Σ but divides it inside the square root.

8 What is and is not shown

- The $(0, 1]$ support is *excluded* by the $J = 0$ data at $\geq 3.3\sigma$ (48I) and $\geq 9\sigma$ (64I), while $[-1, 1]$ is compatible. This conclusion does not depend on any value of $|V_{us}|$.

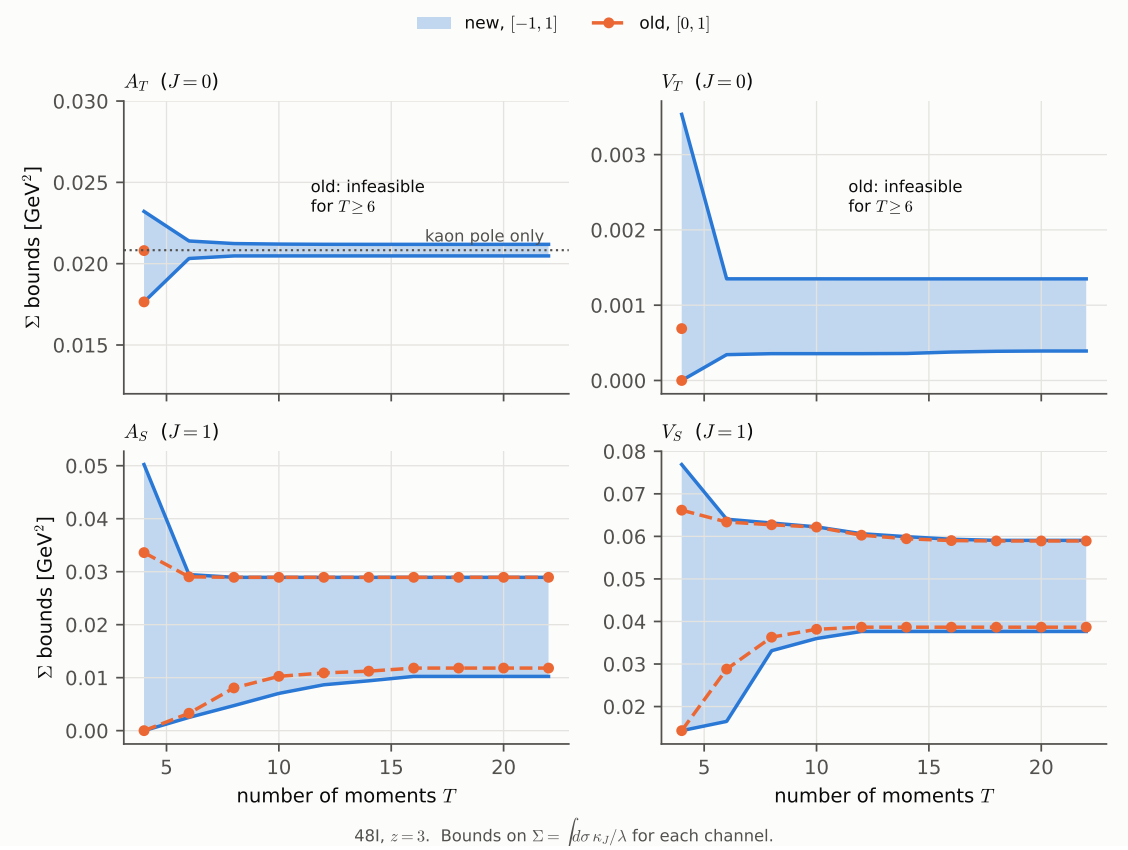


Figure 3: Bounds on $\hat{\Sigma}$ versus the number of moments T (48I, $z = 3$). New support: blue band. Old support: orange dashed where feasible; it is infeasible for A_T and V_T from $T = 6$ on. Beyond $T \approx 18$ the relative error of $C(t)$ exceeds 25% and the bounds stop improving. The $J = 1$ axial channel is the loosest.

- The $|V_{us}|$ interval is wide ($\approx \pm 11\%$). It is a guarantee, not a competitive determination, and the $J = 1$ axial channel limits it.
- The interval is valid only if (i) $\sigma \geq 0$ on $[-1, 1]$ holds for this action (tested at odd t_s, not proven), (ii) the lowest state is the kaon, (iii) $\mathcal{B}_s$ (HFLAV 2023), Z, a^{-1} and M_0 are correct, and (iv) the Gaussian error model holds. With $z = 3$ and $T = 18$ the statistical confidence is at least 95%.
- Not included: the continuum extrapolation (each ensemble has a single lattice spacing), the errors of a^{-1} ($\approx 0.2\%$ on $|V_{us}|$) and of Z, the full covariance (boxes are used instead), lattice artefacts of the local currents, and finite-volume effects beyond the positivity of the measure.
- Next steps: an exact free-field Möbius check at $M_5 = 1.8$, where a single negative pivot would be decisive; a sum-of-squares (SDP) implementation with an exact, grid-free certificate; and the full covariance as an ellipsoid constraint, which would tighten the bounds.

A Reproducing the numbers

All code is in `hausdorff/` of the repository. Run it from the directory that contains `RBCQCD/`.

ensemble	support	$\Sigma_{\rm tot}$ [GeV^2]	$\lvert V_{us}\rvert$	width
48I	old $[0,1]$	infeasible	—	—
64I	old $[0,1]$	infeasible	—	—
48I	new $[-1,1]$	[0.0687, 0.1105]	[0.1962, 0.2489]	11.8%
64I	new $[-1,1]$	[0.0636, 0.1089]	[0.1977, 0.2588]	13.4%
both ensembles, overlap ($z=3$)			**[0.1977, 0.2488]**	
with the $\mathcal{B}_s$ error ($\pm1\sigma$) added			[0.1961, 0.2509]	
both ensembles, overlap ($z=2$)			[0.1998, 0.2351]	

Table 4: $T = 18$, $\mathcal{B}_s = 0.02908(48)$. Since $|V_{us}| \propto \sqrt{\mathcal{B}_s}$, the $\mathcal{B}_s$ error shifts the interval ends by $\mp 0.8\%$. The 48I and 64I intervals are intersected, not combined in a fit. The value $\Sigma_{\rm tot} = 0.0846\,\text{GeV}^2$, which corresponds to $|V_{us}| = 0.2243$ with this $\mathcal{B}_s$, lies inside both intervals. For context (not used as input), HFLAV 2023 quotes $|V_{us}| = 0.2184(21)$ (OPE) and 0.2189(19) (lattice inclusive, Eq. (249)); both lie inside the interval.

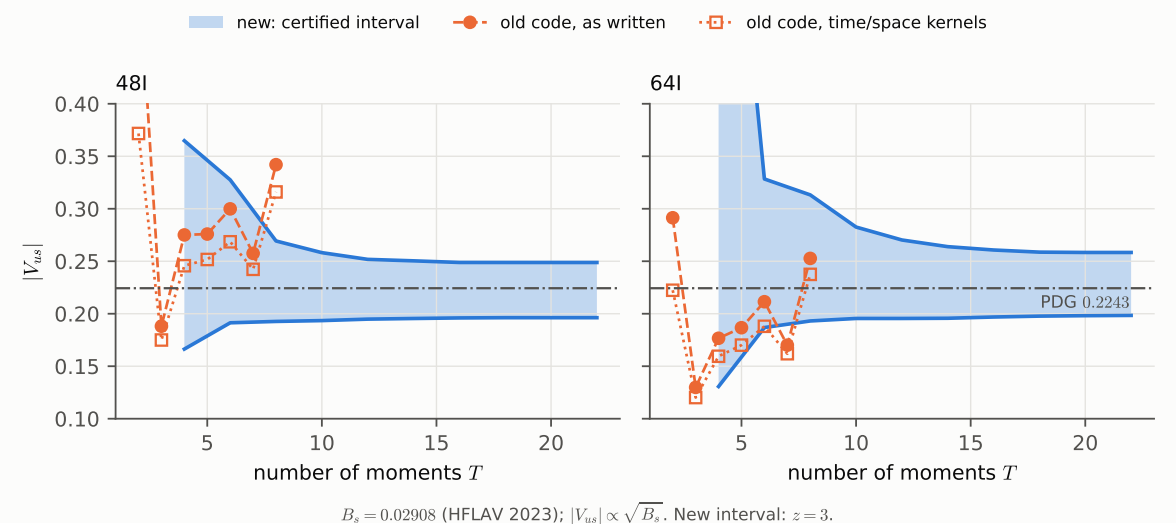


Figure 4: $|V_{us}|$ from the original code (orange; filled: as written, open: time/space kernels) and the certified interval (blue), versus T. $\mathcal{B}_s = 0.02908$ (HFLAV 2023).

`hankel_scan.py`	Hausdorff positivity scan (Table 2)
`cheb_report.py`	$\langle T_n^* \rangle$ and $\langle T_n \rangle$ (Fig. 2)
`bounds_report.py`, `bounds_T.py`	feasibility z^* and bounds versus T
`vus_interval.py`, `vus_certified.py`	channel sums, certificate check, $\lvert V_{us}\rvert$ interval (`BS=...` sets $\mathcal{B}_s$)
`safe_vus.py`	library and single-state calibration (`python safe_vus.py`)
`old_code_check/`	the original `AllClass.py` run without `cupy`
`paper/make_figs.py`	all figures, including the certificates of Fig. 1

Parameters: 48I: $a^{-1} = 1.7295\,\text{GeV}$, $Z = 0.71076$, $M_0 = 0.28816$, $L = 48$, $N_t = 96$, 88 configurations $\times$ 48 sources; 64I: $a^{-1} = 2.3586\,\text{GeV}$, $Z = 0.74293$, $M_0 = 0.21539$, $L = 64$, $N_t = 128$, 40 $\times$ 64. The 64I split is inferred from the row count; the Hankel scan gives identical verdicts for 40×64 and 80×32.

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